

Wenu: 77 stellar-designation cases

Individual evidence review • 6 October 2026

Repository baseline: main 04cc3c1aab5e88a9c5d51f56cb07eada62f4951e

The 77 discrepancies are not 77 errors. Historical sharing, numbering traditions and system/component scope explain much of the evidence. The individual assessments also identify probable misidentifications, coverage gaps and unresolved cross-index problems.

Every case is assessed below. All 77 Fernando decisions remain pending. No catalogue preference, repository content or chart behaviour was changed.

Primary classification	Cases
Historical shared constellation	12
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Identity or cross-index problem	4
Latin-letter case problem	1

Especially valuable historical material

Alpheratz and Elnath's shared Bayer designations; duplicate Flamsteed numbering; δ Eri / ν^3 Eri; κ / κ Pup; ν / ν Ara; ϕ Hya / 2 Crt; ψ Dra numbering; and the documented correction of 1 / 11 Sgr.

Report priority and confidence are different: a high-priority unresolved cross-identification is not automatically ready for publication. The HTML dossier offers searching, classification filters and expandable identity evidence. The JSON preserves structured records for a later implementation.

Method and limits

- This research classifies the designation disagreements. It is not a comprehensive astrophysical review of every object.
- Every case was individually checked against the original ledger, catalogue rows/notes and a live SIMBAD identity. Historical depth varies: not every exponent has an established first author or date.
- Catalogue omission is not negative evidence of historical existence. BSC main name fields omit some Latin letters; Wikidata and WGSN have selective coverage.
- HYG and BSC agreement must not be counted as two independent historical witnesses without tracing HYG upstream provenance.
- Superscript numerals can distinguish unrelated stars, components or catalogue numbering traditions. They are not interchangeable with A/B component letters.
- The κ CrA pair has conflicting HD cross-identifications. The apparent Bayer disagreement becomes a cross-index issue when coordinates are compared; no source was repaired.
- Research recommendations are distinct from Fernando's decisions. All 77 ledger statuses remain pending, all decision/Selman fields null, and no repository or chart behaviour was changed.

Method

The repository's audited source bytes were checked against their recorded hashes. Every case was individually checked against catalogue rows, alternatives and BSC notes, and its HIP identity queried in SIMBAD. Targeted historical evidence came from original Bayer tables, Hoffleit's 1979 study, Wagman's 1987 study and a modern independent reconstruction of Flamsteed identities.

This is a designation review, not a complete astrophysical monograph. Many superscripts still lack a traced first author or adoption date; those questions are explicit. Modern measurements and physical associations should receive a separate research pass before appearing in an observing report.

01. HIP 677 — α And

Historical shared constellation • Report priority: high • Classification confidence: high

Audited source	bayer evidence
Kostjuk	α And
BSC	α And
HYG	α And
WGSN	α And
Wikidata	α And, δ Peg

Assessment. Alpheratz is a genuine shared star, not a mistaken extra designation. In Bayer's Pegasus table (Tabula XIX, facsimile leaf 67), δ describes the navel shared with Andromeda's head. The Andromeda table (Tabula XX, leaf 70) assigns α to that same head. BSC HR 15 also records the former δ Peg designation; current SIMBAD retains both α And and δ Peg.

Retention / reporting recommendation. Preserve δ Peg as a historical alias alongside the modern Andromeda identity. A chart note can explain the shared corner of the Great Square.

Open question or limit. The identity and historical sharing are established. Date-specific details of later abandonment require an edition-by-edition history.

Current SIMBAD: * alf And; ICRS 2.09691619°, 29.09043112°. Kostjuk HD: 358; HR: 15. WGSN name: Alpheratz.

Evidence: Bayer: Tabula XIX and XX, facsimile leaves 67–71; BSC HR 15 name note.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [bayer](#). Full references follow.

02. HIP 8832 — γ 1 Ari

System and component scope • Report priority: high • Classification confidence: high

Audited source	bayer evidence
Kostjuk	γ 1 Ari
BSC	γ 2 Ari
HYG	γ 2 Ari
WGSN	No value in the audited comparison
Wikidata	γ Ari

Assessment. Mesarthim exposes both a component-number disagreement and a system-level HIP entry. Kostjuk explicitly records γ 1 and γ 2 alternatives for HD 11502, with different source references. BSC HR 546 is a component of the HR 545–546 pair, while the current SIMBAD HIP 8832 object is γ Ari AB. Its unsuffixed designation therefore does not resolve the component numbering.

Retention / reporting recommendation. Retain the alternatives with their component and catalogue provenance. Explain the visual double in a report; do not let a majority vote assign one component's suffix to the HIP system.

Open question or limit. A preferred component mapping needs a coordinate and double-star crosswalk; γ 1 and γ 2 must not become interchangeable component IDs.

Current SIMBAD: * gam Ari; ICRS 28.38256161°, 19.29385527°. Kostjuk HD: 11502; HR: 546.

Evidence: Kostjuk table1 HD 11502; BSC HR 545–546; SIMBAD HIP 8832.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

03. HIP 9347 — υ Cet

Superscript convention • Report priority: low • Classification confidence: moderate

Audited source	bayer evidence
Kostjuk	υ Cet
BSC	υ Cet
HYG	υ Cet
WGSN	No value in the audited comparison
Wikidata	υ1 Cet

Assessment. HIP 9347 is 59 Cet / HD 12274. Kostjuk, BSC, HYG and current SIMBAD use bare υ Cet. The ledger's υ1 Cet occurs only in Wikidata; the examined Kostjuk alternative table supplies no independent support for the exponent. Hoffleit's historical discussion of 59 Cet shows that older numbering existed, but does not establish this particular υ1 assertion.

Retention / reporting recommendation. Keep the raw Wikidata assertion for review and the bare designation as documented evidence. This is a provenance question, not a report-ready historical story about υ1.

Open question or limit. Locate the specific source behind the υ1 Wikidata statement before presenting that form as an established historical alias.

Current SIMBAD: * ups Cet; ICRS 30.00128530°, -21.07783167°. Kostjuk HD: 12274; HR: 585.

Evidence: Hoffleit 1979, Cetus discussion, printed p.48; BSC HR 585.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hoffleit](#). Full references follow.

04. HIP 9640 — γ1 And

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	γ1 And
BSC	γ1 And
HYG	γ1 And
WGSN	γ And
Wikidata	γ And

Assessment. Almach's γ1 And is the bright component, HD 12533 / HR 603. The unsuffixed γ And also identifies it in SIMBAD and is used in the WGSN name table. BSC distinguishes the neighbouring HR 604 component; Kostjuk preserves both suffixed and unsuffixed traditions and even records a γ2 alternative in one source.

Retention / reporting recommendation. Report Almach as a multiple-star object and preserve the component-qualified label where required. Treat the unsuffixed form as context-dependent, not

a different star.

Open question or limit. Do not transfer the proper name or component suffix to every star in the system without checking its target.

Current SIMBAD: * gam01 And; ICRS 30.97480121°, 42.32972842°. Kostjuk HD: 12533; HR: 603. WGSN name: Almach.

Evidence: Kostjuk table1 HD 12533; BSC HR 603-604; WGSN Almach row.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

05. HIP 12832 — 38 Ari

Historical shared constellation • Report priority: high • Classification confidence: high

Audited source	flam evidence
Kostjuk	38 Ari
BSC	38 Ari
HYG	38 Ari
WGSN	No value in the audited comparison
Wikidata	38 Ari, 88 Cet

Assessment. 38 Ari and 88 Cet refer to the same HIP 12832 / HD 17093. The Kostjuk addendum explicitly marks the two entries as a duplicate; Wagman's historical catalogue study corroborates the pairing. Current SIMBAD retains both identifiers. The alternative constellation number is historical information rather than evidence for a second plotted object.

Retention / reporting recommendation. Preserve 88 Cet as a historical catalogue alias. A report can explain that older constellational lists could enumerate the same star twice.

Open question or limit. The duplicate is established; do not attribute it to a particular modern boundary change without checking the original catalogue entry.

Current SIMBAD: * 38 Ari; ICRS 41.23992264°, 12.44575607°. Kostjuk HD: 17093; HR: 812.

Evidence: Kostjuk addendum HD 17093; Wagman 1987, Cetus, printed p.215; Flamsteed reconstruction.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wagman](#); [flamsteed](#). Full references follow.

06. HIP 12843 — 1 Eri

Historical shared constellation • Report priority: high • Classification confidence: high

Audited source	flam evidence
Kostjuk	1 Eri
BSC	1 Eri
HYG	1 Eri
WGSN	No value in the audited comparison
Wikidata	1 Eri, 90 Cet

Assessment. 1 Eri, τ 1 Eri and 90 Cet identify HIP 12843 / HD 17206. The addendum pairs 90 Cet with τ 1 Eri; SIMBAD independently retains 1 Eri, τ 1 Eri and 90 Cet. Thus the three expressions

combine a Flamsteed number, a Bayer-style designation and a historical duplicate number.

Retention / reporting recommendation. Preserve the complete alias set with each designation's type. Use a report note about the Cetus-Eridanus overlap rather than treating 90 Cet as an erroneous modern location.

Open question or limit. The exact catalogue chronology of numbering the τ group needs original-edition citations if expanded into a longer historical narrative.

Current SIMBAD: * tau01 Eri; ICRS 41.27582834°, -18.57262283°. Kostjuk HD: 17206; HR: 818.

Evidence: Kostjuk addendum HD 17206; Wagman 1987, Cetus, printed p.215.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wagman](#). Full references follow.

07. HIP 13702 — ρ Ari

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ρ Ari
BSC	ρ_3 Ari
HYG	ρ_3 Ari
WGSN	No value in the audited comparison
Wikidata	ρ_3 Ari

Assessment. 46 Ari / HD 18256 is represented as both ρ Ari and ρ_3 Ari in the Kostjuk alternative table. BSC and current SIMBAD recognize the suffixed form; SIMBAD also retains bare ρ . The disagreement is therefore documented abbreviation or numbering practice, rather than a cross-identification failure.

Retention / reporting recommendation. Retain both strings, with ρ_3 available when distinguishing the historical ρ group. A short nomenclature note is sufficient.

Open question or limit. A suffix alone does not establish physical association with the other numbered ρ stars.

Current SIMBAD: * rho Ari; ICRS 44.10904886°, 18.02324033°. Kostjuk HD: 18256; HR: 869.

Evidence: Kostjuk table1 HD 18256; BSC HR 869.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

08. HIP 13847 — θ 1 Eri

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	θ 1 Eri
BSC	θ 1 Eri
HYG	θ 1 Eri
WGSN	θ 1 Eri
Wikidata	θ Eri

Assessment. Acamar's HIP entry resolves to θ Eri AB in current SIMBAD. The source rows use $\theta 1$ for HD 18622 / HR 897 and distinguish HR 898; BSC notes combined light. The bare Wikidata θ Eri can describe the pair, while $\theta 1$ describes a component. Kostjuk's alternatives also include both suffixes for the same HD identification.

Retention / reporting recommendation. Keep system and component aliases separately. A report may explain that a single chart point represents a double star.

Open question or limit. Check the precise WGSN name target before assigning Acamar to both components; the Bayer string alone is insufficient.

Current SIMBAD: * tet Eri; ICRS 44.56531356°, -40.30468123°. Kostjuk HD: 18622; HR: 897. WGSN name: Acamar.

Evidence: Kostjuk table1 HD 18622; BSC HR 897-898; SIMBAD HIP 13847; WGSN Acamar row.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

09. HIP 15457 — κ Cet

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	κ Cet
BSC	$\kappa 1$ Cet
HYG	$\kappa 1$ Cet
WGSN	No value in the audited comparison
Wikidata	$\kappa 1$ Cet

Assessment. 96 Cet / HD 20630 has κ Cet and $\kappa 1$ Cet explicitly documented in Kostjuk's alternatives. Current SIMBAD retains both, and BSC's $\kappa 1$ row matches the same star. There is no evidence here that the bare form selects a different HIP object.

Retention / reporting recommendation. Preserve the unsuffixed and numbered forms as source-qualified aliases. The disagreement need not become a default interesting-object entry.

Open question or limit. Historical use of the exponent can be traced further if an expanded nomenclature report is desired.

Current SIMBAD: * kap01 Cet; ICRS 49.84040174°, 3.37019861°. Kostjuk HD: 20630; HR: 996.

Evidence: Kostjuk table1 HD 20630; BSC HR 996.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

10. HIP 15627 — τ Ari

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	τ Ari
BSC	$\tau 1$ Ari
HYG	$\tau 1$ Ari

WGSN	No value in the audited comparison
Wikidata	τ1 Ari

Assessment. 61 Ari / HD 20756 is τ Ari in several source traditions and τ1 Ari in BSC and current SIMBAD. Kostjuk records both explicitly. BSC's old suspected-double note is separate evidence and does not by itself explain why the τ exponent was introduced.

Retention / reporting recommendation. Retain both forms; avoid claiming the suffix means a binary component on the strength of this case alone.

Open question or limit. The first historical introduction of the exponent remains untraced.

Current SIMBAD: * tau01 Ari; ICRS 50.30676927°, 21.14706638°. Kostjuk HD: 20756; HR: 1005.

Evidence: Kostjuk table1 HD 20756; BSC HR 1005.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

11. HIP 19990 — ω Tau

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ω Tau
BSC	ω2 Tau
HYG	ω2 Tau
WGSN	No value in the audited comparison
Wikidata	ω2 Tau

Assessment. 50 Tau / HD 27045 is ω Tau in Kostjuk and ω2 Tau in BSC. Kostjuk's alternative table documents both traditions, and current SIMBAD retains both. A numbering distinction within a letter group is sufficient to explain the ledger difference.

Retention / reporting recommendation. Keep both aliases and use a numbered form when the letter group needs disambiguation.

Open question or limit. Do not describe the ω stars as a bound multiple system without separate evidence.

Current SIMBAD: * ome02 Tau; ICRS 64.31525469°, 20.57859738°. Kostjuk HD: 27045; HR: 1329.

Evidence: Kostjuk table1 HD 27045; BSC HR 1329.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

12. HIP 20455 — δ Tau

Superscript convention • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	δ Tau
BSC	δ1 Tau
HYG	δ1 Tau

WGSN	δ1 Tau
Wikidata	δ1 Tau

Assessment. The Hyades star Secunda Hyadum, 61 Tau / HD 27697, has both δ Tau and δ1 Tau in historical catalogue use. BSC, WGSN and SIMBAD connect the numbered form to this same object; Kostjuk preserves the unnumbered tradition. This is not a different Hyades member accidentally joined to the HIP row.

Retention / reporting recommendation. Preserve the two Bayer forms and the proper name. A Hyades-oriented chart report could include the naming context, independently of the audit discrepancy.

Open question or limit. A narrative about the whole δ group should distinguish cluster membership from component nomenclature.

Current SIMBAD: * del Tau; ICRS 65.73372983°, 17.54251383°. Kostjuk HD: 27697; HR: 1373. WGSN name: Secunda Hyadum.

Evidence: Kostjuk table1 HD 27697; BSC HR 1373; WGSN Secunda Hyadum row.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

13. HIP 20535 — d Eri

Competing letter traditions • Report priority: high • Classification confidence: high

Audited source	bayer evidence
Kostjuk	d Eri
BSC	No value in the audited comparison
HYG	No value in the audited comparison
WGSN	υ3 Eri
Wikidata	d Eri, υ3 Eri

Assessment. HIP 20535 / HR 1393 is 43 Eri, called d Eri in one catalogue tradition and υ3 Eri in another. Hoffleit's Eridanus investigation traces multiple historical d assignments; consequently d Eri is not a globally unambiguous synonym. Kostjuk records both forms for HD 28028. SIMBAD and WGSN connect this HIP object with υ3 Eri and Beemim.

Retention / reporting recommendation. Preserve d Eri as a historically qualified alias of this specific HIP, and explain the competing tradition when requested. Never resolve a bare d Eri query by text alone.

Open question or limit. An original-atlas reconstruction of all d assignments would be needed for a detailed chronology.

Current SIMBAD: * 43 Eri; ICRS 66.00919306°, -34.01684951°. Kostjuk HD: 28028; HR: 1393. WGSN name: Beemim.

Evidence: Hoffleit 1979, Eridanus, printed pp.50–51; Kostjuk table1 HD 28028; WGSN Beemim row.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hoffleit](#); [wgsn](#). Full references follow.

14. HIP 20635 — κ Tau

Superscript convention • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	κ Tau
BSC	κ1 Tau
HYG	κ1 Tau
WGSN	No value in the audited comparison
Wikidata	κ1 Tau

Assessment. 65 Tau / HD 27934 is κ Tau or κ1 Tau according to catalogue convention; both are in Kostjuk's alternative table. BSC distinguishes κ2 Tau as HR 1388 and discusses multiplicity. Current SIMBAD identifies this HIP with κ1, so bare κ does not erase the nearby separately catalogued κ2 object.

Retention / reporting recommendation. Preserve the bare alias but keep κ1 specific to HIP 20635. A report can discuss the nearby pair if both objects are in the chart.

Open question or limit. Component letters inside each κ star must be distinguished from the superscript numbering of the wider pair.

Current SIMBAD: * kap01 Tau; ICRS 66.34235197°, 22.29387109°. Kostjuk HD: 27934; HR: 1387.

Evidence: Kostjuk table1 HD 27934; BSC HR 1387–1388.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

15. HIP 21589 — c1 Tau

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	c1 Tau
BSC	No value in the audited comparison
HYG	No value in the audited comparison
WGSN	No value in the audited comparison
Wikidata	c Tau

Assessment. 90 Tau / HD 29388 has Latin c and c1 alternatives, explicitly recorded by Kostjuk. BSC's main name field gives the Flamsteed designation rather than this Latin label; its empty Bayer extraction is not evidence against the alias. Current SIMBAD retains c Tau.

Retention / reporting recommendation. Preserve the Latin letter and case, and retain the exponent as a source-specific variant. This is mainly catalogue archaeology.

Open question or limit. The origin of the c1 exponent is not established by the compared rows alone.

Current SIMBAD: * c Tau; ICRS 69.53942456°, 12.51083696°. Kostjuk HD: 29388; HR: 1473.

Evidence: Kostjuk table1 HD 29388; BSC HR 1473; Hoffleit 1979, Taurus, printed p.64.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hoffleit](#). Full references follow.

16. HIP 23595 — γ Cae

Superscript convention • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	γ Cae
BSC	$\gamma 1$ Cae
HYG	$\gamma 1$ Cae
WGSN	No value in the audited comparison
Wikidata	$\gamma 1$ Cae

Assessment. HIP 23595 / HD 32831 is $\gamma 1$ Cae, also called γ Cae. Kostjuk documents both; SIMBAD identifies the object with the unsuffixed main name and the $\gamma 1$ alias. It must be distinguished from HIP 23596 / $\gamma 2$ Cae, a separate nearby HIP entry.

Retention / reporting recommendation. Preserve the bare alias but make the paired $\gamma 1/\gamma 2$ distinction available for charts containing both stars.

Open question or limit. Nearby numbered stars are not automatically components of a bound system.

Current SIMBAD: * gam Cae; ICRS 76.10167558°, -35.48297537°. Kostjuk HD: 32831; HR: 1652.

Evidence: Kostjuk table1 HD 32831; BSC HR 1652; SIMBAD HIP 23595–23596.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

17. HIP 23596 — γ Cae

Superscript convention • Report priority: medium • Classification confidence: moderate

Audited source	bayer evidence
Kostjuk	γ Cae
BSC	$\gamma 2$ Cae
HYG	$\gamma 2$ Cae
WGSN	No value in the audited comparison
Wikidata	$\gamma 2$ Cae

Assessment. HIP 23596 / HD 32846 is $\gamma 2$ Cae in BSC and SIMBAD; its variable-star identification is X Cae. Kostjuk's main row omits the exponent and supplies no alternative-table explanation for this HD. The existence of the separate $\gamma 1$ HIP 23595 makes the bare γ ambiguous when both appear.

Retention / reporting recommendation. Keep $\gamma 2$ specific to this HIP and record γ as the source's coarse label. A report can explain the paired designation without implying a physical pair.

Open question or limit. The reason for Kostjuk's unsuffixed row needs a direct source trace; it is not independently vindicated by an alternative entry.

Current SIMBAD: * gam02 Cae; ICRS 76.10898791°, -35.70510624°. Kostjuk HD: 32846; HR: 1653.

Evidence: BSC HR 1653; SIMBAD HIP 23596.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

18. HIP 25428 — β Tau

Historical shared constellation • Report priority: high • Classification confidence: high

Audited source	bayer evidence
Kostjuk	β Tau
BSC	β Tau
HYG	β Tau
WGSN	β Tau
Wikidata	β Tau, γ Aur

Assessment. Elnath has a genuine β Tau / γ Aur history. Bayer's Auriga table, Tabula XII (facsimile leaf 46), places γ at the figure's ankle and Taurus's northern horn. BSC HR 1791 and the Kostjuk duplicate addendum preserve the overlap; SIMBAD retains β Tau, γ Aur, 112 Tau and 23 Aur.

Retention / reporting recommendation. Preserve γ Aur as a historical alias. Pair this case with Alpheratz as an especially clear report on shared constellation figures.

Open question or limit. A modern label should reflect the selected modern constellation, while the historical association remains separately reportable.

Current SIMBAD: * bet Tau; ICRS 81.57297133°, 28.60745172°. Kostjuk HD: 35497; HR: 1791. WGSN name: Elnath.

Evidence: Bayer Tabula XII, facsimile leaf 46; BSC HR 1791; Kostjuk addendum HD 35497.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [bayer](#). Full references follow.

19. HIP 25473 — ψ Ori

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ψ Ori
BSC	ψ 2 Ori
HYG	ψ 2 Ori
WGSN	No value in the audited comparison
Wikidata	ψ 2 Ori

Assessment. 30 Ori / HD 35715 has ψ Ori and ψ 2 Ori in Kostjuk's alternatives and current SIMBAD. The comparison represents a numbered-letter convention for the same HIP object. The old BSC component discussion is not a reason to equate ψ 1 and ψ 2 with A and B automatically.

Retention / reporting recommendation. Retain both aliases; preserve ψ 2 when disambiguating Orion's ψ designations.

Open question or limit. Any physical account of the star's companions requires separate modern evidence.

Current SIMBAD: * psi02 Ori; ICRS 81.70928885°, 3.09567282°. Kostjuk HD: 35715; HR: 1811.

Evidence: Kostjuk table1 HD 35715; BSC HR 1811.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

20. HIP 27949 — 30 Aur

Historical shared constellation • Report priority: high • Classification confidence: high

Audited source	flam evidence
Kostjuk	30 Aur
BSC	30 Aur
HYG	30 Aur
WGSN	No value in the audited comparison
Wikidata	30 Aur, 32 Cam

Assessment. 30 Aur / ξ Aur and 32 Cam identify HIP 27949 / HD 39283. BSC explicitly records the Flamsteed double identification, and the Kostjuk addendum pairs 32 Cam with ξ Aur. SIMBAD retains both number designations.

Retention / reporting recommendation. Preserve 32 Cam as a historical alias and explain the duplicate numbering in an optional report.

Open question or limit. Do not assume that every obsolete cross-constellation number resulted from the 1930 boundaries; this duplicate is already a catalogue history.

Current SIMBAD: * ksi Aur; ICRS 88.71152891°, 55.70696671°. Kostjuk HD: 39283; HR: 2029.

Evidence: BSC HR 2029 name note; Kostjuk table1 and addendum HD 39283; Flamsteed reconstruction.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [flamsteed](#). Full references follow.

21. HIP 36363 — y Pup

Superscript convention • Report priority: medium • Classification confidence: moderate

Audited source	bayer evidence
Kostjuk	y Pup
BSC	No value in the audited comparison
HYG	No value in the audited comparison
WGSN	No value in the audited comparison
Wikidata	y1 Pup

Assessment. HIP 36363 / HD 59635 is Latin y Pup in Kostjuk and SIMBAD, with y1 in the Becvar-linked alternative row. Hoffleit's Puppis study separates several historically confused Latin Y/y labels and the variable Y Pup. It warns against collapsing case and superscripts in this constellation.

Retention / reporting recommendation. Preserve y1 as a qualified historical variant and y as the current database alias. Keep this HIP distinct from the variable Y Pup and other historical Y assignments.

Open question or limit. The earliest source for this particular exponent was not securely established by Hoffleit; further attribution should remain provisional.

Current SIMBAD: * y Pup; ICRS 112.27372998°, -38.81206732°. Kostjuk HD: 59635; HR: 2875.

Evidence: Kostjuk table1 HD 59635; Hoffleit 1979, Puppis, printed pp.58-59.

22. HIP 37229 — κ 1 Pup

Competing letter traditions • Report priority: high • Classification confidence: high

Audited source	bayer evidence
Kostjuk	κ 1 Pup
BSC	No value in the audited comparison
HYG	No value in the audited comparison
WGSN	No value in the audited comparison
Wikidata	k Pup

Assessment. HIP 37229 represents the k Pup double in SIMBAD. Kostjuk records κ 1, κ , k and another κ exponent for HD 61555. BSC's notes describe several k/ κ usages involving HR 2873, 2881, 2948 and 2949. Hoffleit traces differing atlas and catalogue letter traditions; this is a documented historical ambiguity, not merely Unicode corruption.

Retention / reporting recommendation. Retain Greek κ and Latin k as distinct, provenance-bearing strings. An interesting-object report should explain why an old κ Pup label can refer to this system.

Open question or limit. A unique default display form and component mapping still need Fernando's decision; do not silently normalize κ to k.

Current SIMBAD: * k Pup; ICRS 114.70779496°, -26.80383873°. Kostjuk HD: 61555; HR: 2948.

Evidence: Hoffleit 1979, Puppis, printed pp.60–61; BSC HR 2948–2949; Kostjuk table1 HD 61555.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hoffleit](#). Full references follow.

23. HIP 39191 — ω Cnc

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ω Cnc
BSC	ω 1 Cnc
HYG	ω 1 Cnc
WGSN	No value in the audited comparison
Wikidata	ω 1 Cnc

Assessment. 2 Cnc / HD 65714 is ω Cnc or ω 1 Cnc in the examined catalogues. Kostjuk documents both; SIMBAD includes both identifiers on the same object. This is a numbered-letter distinction rather than evidence for two objects at HIP 39191.

Retention / reporting recommendation. Preserve the aliases; use the exponent only as an explicit display convention.

Open question or limit. The numbering does not establish a physical connection to other ω stars.

Current SIMBAD: * ome Cnc; ICRS 120.23280389°, 25.39283739°. Kostjuk HD: 65714; HR: 3124.

Evidence: Kostjuk table1 HD 65714; BSC HR 3124.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

24. HIP 39780 — μ Cnc

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	μ Cnc
BSC	$\mu 2$ Cnc
HYG	$\mu 2$ Cnc
WGSN	No value in the audited comparison
Wikidata	$\mu 2$ Cnc

Assessment. 10 Cnc / HD 67228 has μ Cnc and $\mu 2$ Cnc explicitly documented by Kostjuk, with both retained by SIMBAD. Its source rows agree on the physical identity despite different label resolution.

Retention / reporting recommendation. Retain both forms with source attribution; no automatic error correction is justified.

Open question or limit. A detailed account of the μ numbering would require earlier atlas or catalogue entries.

Current SIMBAD: * μ .02 Cnc; ICRS 121.94106771°, 21.58181300°. Kostjuk HD: 67228; HR: 3176.

Evidence: Kostjuk table1 HD 67228; BSC HR 3176.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

25. HIP 39953 — $\gamma 2$ Vel

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	$\gamma 2$ Vel
BSC	$\gamma 2$ Vel
HYG	$\gamma 2$ Vel
WGSN	No value in the audited comparison
Wikidata	γ Vel

Assessment. HIP 39953 / HD 68273 is $\gamma 2$ Vel, also known under the bare γ Vel system label. Current SIMBAD classifies the object as a Wolf-Rayet star, and BSC records blended companions. Kostjuk preserves bare γ and several additional letter/suffix discrepancies, showing that the full evidence is richer than the ledger's two displayed values.

Retention / reporting recommendation. Preserve $\gamma 2$ and the bare system alias separately. This is a candidate for an astrophysical report after a dedicated modern-source review of the Wolf-Rayet binary.

Open question or limit. Do not publish BSC's old orbital or companion measurements as current values; $\gamma 1$ Vel must remain separately identifiable.

Current SIMBAD: * gam02 Vel; ICRS 122.38312556°, -47.33658633°. Kostjuk HD: 68273; HR: 3207.

Evidence: Kostjuk table1 HD 68273; BSC HR 3206–3207; SIMBAD HIP 39953.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

26. HIP 40023 — ψ Cnc

Superscript convention • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ψ Cnc
BSC	ψ Cnc
HYG	ψ Cnc
WGSN	No value in the audited comparison
Wikidata	$\psi 2$ Cnc

Assessment. 14 Cnc / HD 67767 is usually ψ Cnc, but BSC explicitly also calls it $\psi 2$ and identifies $\psi 1$ with 13 Cnc / HD 67690. Thus Wikidata's suffix has independent catalogue support even though the BSC main row is unsuffixed. This is a concrete example of why comparing main fields alone overstates disagreement.

Retention / reporting recommendation. Preserve $\psi 2$ as a disambiguating alias. A brief report note can point out the fainter $\psi 1$ counterpart.

Open question or limit. The relation of these stars is nomenclatural here; no physical association is inferred.

Current SIMBAD: * psi Cnc; ICRS 122.61325359°, 25.50733393°. Kostjuk HD: 67767; HR: 3191.

Evidence: BSC HR 3191 name note; Kostjuk table1 HD 67767.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

27. HIP 40167 — $\zeta 1$ Cnc / $\zeta 2$ Cnc

System and component scope • Report priority: high • Classification confidence: high

Audited source	bayer evidence
Kostjuk	$\zeta 1$ Cnc, $\zeta 2$ Cnc
BSC	$\zeta 2$ Cnc
HYG	$\zeta 2$ Cnc
WGSN	ζ Cnc
Wikidata	ζ Cnc

Assessment. HIP 40167 is the ζ Cnc multiple, not a single unambiguously numbered component. Kostjuk has two HD rows under the HIP; BSC distinguishes HR 3208, 3209 and 3210, while the ledger's BSC value comes from HR 3210. WGSN's Tegmine row targets HR 3208, and SIMBAD's HIP object is AB,C. The compared sources therefore do not all select the same subobject.

Retention / reporting recommendation. Retain ζ, ζ1 and ζ2 with explicit targets. A report can describe the multiple system, but the chart label and named component must remain distinct.

Open question or limit. Build the full HD–HR–component crosswalk before choosing a canonical suffix or distributing Tegmine across the system.

Current SIMBAD: * zet Cnc; ICRS 123.05302654°, 17.64776708°. Kostjuk HD: 68256, 68257; HR: 3208, 3209. WGSN name: Tegmine.

Evidence: Kostjuk table1 HD 68256–68257; BSC HR 3208–3210; WGSN Tegmine row; SIMBAD HIP 40167.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

28. HIP 45410 — π Cnc

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	π Cnc
BSC	π2 Cnc
HYG	π2 Cnc
WGSN	No value in the audited comparison
Wikidata	π2 Cnc

Assessment. 82 Cnc / HD 79554 has π Cnc and π2 Cnc in Kostjuk and SIMBAD. BSC specifically notes that it is usually called simply π Cnc. The more detailed number is compatible with the commonly abbreviated form.

Retention / reporting recommendation. Preserve both aliases; this does not require a prominent historical note by default.

Open question or limit. The first introduction of π2 is untraced in this review.

Current SIMBAD: * pi.02 Cnc; ICRS 138.80771714°, 14.94150888°. Kostjuk HD: 79554; HR: 3669.

Evidence: BSC HR 3669 name note; Kostjuk table1 HD 79554.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

29. HIP 48552 — 9 Sex

Identity or cross-index problem • Report priority: high • Classification confidence: high

Audited source	flam evidence
Kostjuk	9 Sex
BSC	No value in the audited comparison
HYG	8 Sex
WGSN	No value in the audited comparison
Wikidata	9 Sex

Assessment. HIP 48552 / HD 85762 is 9 Sex in Kostjuk, current SIMBAD and the reconstructed Flamsteed catalogue. The HYG 8 Sex conflicts with a distinct object: 8 Sex is γ Sex / HD 85558 / HIP 48437. Unlike the shared-constellation duplicates, the examined evidence does not support 8 and

9 Sex as aliases of one star.

Retention / reporting recommendation. Flag HYG's mapping as a likely identification error. Preserve its raw value as rejected-or-disputed evidence, not as an eligible historical alias.

Open question or limit. Trace the HYG row's upstream origin before assigning the exact failure mechanism; a typo and a join error are different explanations.

Current SIMBAD: * 9 Sex; ICRS 148.52793419°, 4.94541565°. Kostjuk HD: 85762; HR: .

Evidence: Flamsteed reconstruction, Sextans 8 and 9; SIMBAD HIP 48552 and HIP 48437.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [flamsteed](#). Full references follow.

30. HIP 49065 — μ Cha

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	μ Cha
BSC	$\mu 1$ Cha
HYG	$\mu 1$ Cha
WGSN	No value in the audited comparison
Wikidata	$\mu 1$ Cha

Assessment. HIP 49065 / HD 87971 is μ Cha in Kostjuk and current SIMBAD, but $\mu 1$ Cha in BSC. Kostjuk's alternatives explicitly include both. The source's object identity is stable; the suffix is a catalogue convention.

Retention / reporting recommendation. Preserve both forms without treating $\mu 1$ as a correction of a bad HIP association.

Open question or limit. The historical scope of $\mu 1$ versus $\mu 2$ requires separate source work if discussed as a pair.

Current SIMBAD: * μ . Cha; ICRS 150.18247385°, -82.21467013°. Kostjuk HD: 87971; HR: 3983.

Evidence: Kostjuk table1 HD 87971; BSC HR 3983.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

31. HIP 50583 — $\gamma 1$ Leo

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	$\gamma 1$ Leo
BSC	$\gamma 1$ Leo
HYG	$\gamma 1$ Leo
WGSN	$\gamma 1$ Leo
Wikidata	γ Leo

Assessment. Algieba's HIP 50583 is the γ Leo AB object in SIMBAD. The source comparison uses γ 1 for HD 89484 / HR 4057, while BSC identifies HR 4058 as the other component and records blended photometry. Bare γ is appropriate at a different level of object resolution.

Retention / reporting recommendation. Preserve system and component labels; a report can describe the visual double without changing the catalogue preference.

Open question or limit. Current component data and physical measurements need modern sources before inclusion in an observing report.

Current SIMBAD: * gam Leo; ICRS 154.99312733°, 19.84148522°. Kostjuk HD: 89484; HR: 4057. WGSN name: Algieba.

Evidence: Kostjuk table1 HD 89484; BSC HR 4057–4058; WGSN Algieba row.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

32. HIP 52085 — ϕ Hya

Superscript convention • Report priority: high • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ϕ Hya
BSC	ϕ 3 Hya
HYG	ϕ 3 Hya
WGSN	No value in the audited comparison
Wikidata	ϕ 3 Hya

Assessment. HIP 52085 / HD 92214 is ϕ Hya or ϕ 3 Hya; Kostjuk additionally documents ϕ 1 in Becvar. BSC explicitly explains the competing ϕ numbering, and the addendum reveals another valuable alias, 2 Crt. SIMBAD retains ϕ , ϕ 3 and 2 Crt. This case combines historical numbering with a cross-constellation duplicate beyond the original ledger field.

Retention / reporting recommendation. Preserve ϕ 3, bare ϕ and historical 2 Crt; keep the competing ϕ 1 tradition qualified. This deserves an optional historical report entry.

Open question or limit. Do not conflate this star with the distinct faint ϕ 1 Hya / HD 91369.

Current SIMBAD: * phi Hya; ICRS 159.64561531°, -16.87656519°. Kostjuk HD: 92214; HR: 4171.

Evidence: BSC HR 4171 name note; Kostjuk table1 and addendum HD 92214; Hoffleit 1979, printed p.52.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hoffleit](#). Full references follow.

33. HIP 54301 — z Car

Superscript convention • Report priority: low • Classification confidence: moderate

Audited source	bayer evidence
Kostjuk	z Car
BSC	No value in the audited comparison
HYG	No value in the audited comparison
WGSN	No value in the audited comparison
Wikidata	z1 Car

Assessment. HIP 54301 / HD 96566 is Latin z Car in Kostjuk and SIMBAD, with z1 in the Becvar-linked alternative row. BSC's omission of a Latin label from its main name field does not refute it. This is a retained old-letter convention.

Retention / reporting recommendation. Preserve case and the numbered alternative; a specialist nomenclature report may include it.

Open question or limit. The earliest introduction of z1 has not been established.

Current SIMBAD: * z Car; ICRS 166.63511032°, -62.42411406°. Kostjuk HD: 96566; HR: 4325.

Evidence: Kostjuk table1 HD 96566; BSC HR 4325.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

34. HIP 60718 — α1 Cru

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	α1 Cru
BSC	α1 Cru
HYG	α1 Cru
WGSN	α Cru
Wikidata	α Cru

Assessment. Acrux's HIP 60718 corresponds to an AB entry in SIMBAD. BSC's α1 row is HR 4730, with HR 4731 distinguished and combined light noted. WGSN uses the bare α Cru system expression. The ledger mixes a component designation with a system designation.

Retention / reporting recommendation. Keep α1 specific and α Cru at the broader scope. A report can explain what the single bright chart point contains.

Open question or limit. The proper name's target and an observing description of subcomponents should be checked independently of the chart's blended magnitude.

Current SIMBAD: * alf Cru; ICRS 186.64956340°, -63.09909286°. Kostjuk HD: 108248; HR: 4730. WGSN name: Acrux.

Evidence: Kostjuk table1 HD 108248; BSC HR 4730–4731; WGSN Acrux row.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

35. HIP 63462 — 37 Com

Historical shared constellation • Report priority: high • Classification confidence: high

Audited source	flam evidence
Kostjuk	37 Com
BSC	37 Com
HYG	37 Com
WGSN	No value in the audited comparison
Wikidata	13 CVn, 37 Com

Assessment. 37 Com and 13 CVn identify HIP 63462 / HD 112989. The duplicate addendum explicitly pairs them, and current SIMBAD retains both. This is useful historical catalogue information; 13 CVn does not require plotting a second star or locating it in modern Canes Venatici.

Retention / reporting recommendation. Preserve the obsolete number with historical status and a shared event identifier for reporting.

Open question or limit. An extended narrative should consult the original numbered lists to date the duplicate's origin.

Current SIMBAD: * 37 Com; ICRS 195.06862266°, 30.78502207°. Kostjuk HD: 112989; HR: 4924.

Evidence: Kostjuk addendum HD 112989; Flamsteed reconstruction; SIMBAD HIP 63462.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [flamsteed](#). Full references follow.

36. HIP 67627 — 10 Dra

Historical shared constellation • Report priority: high • Classification confidence: high

Audited source	flam evidence
Kostjuk	10 Dra
BSC	10 Dra
HYG	10 Dra
WGSN	No value in the audited comparison
Wikidata	10 Dra, 87 UMa

Assessment. 10 Dra / Latin i Dra and 87 UMa identify HIP 67627 / HD 121130. Wagman explicitly documents 87 UMa as the same star as 10 or i Dra, HR 5226. SIMBAD currently retains 10 Dra and i Dra but not the old 87 UMa alias in the returned identifier set; that absence does not erase the historical evidence.

Retention / reporting recommendation. Preserve 87 UMa as a historical alias supported by the reconstruction, rather than requiring every modern database to retain it.

Open question or limit. The historical pairing is established; exact original observation details remain a possible expansion.

Current SIMBAD: * i Dra; ICRS 207.85796515°, 64.72328128°. Kostjuk HD: 121130; HR: 5226.

Evidence: Wagman 1987, Ursa Major, printed p.223; Flamsteed reconstruction.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wagman](#); [flamsteed](#). Full references follow.

37. HIP 69481 — κ1 Boo

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	κ1 Boo
BSC	κ1 Boo
HYG	κ1 Boo

WGSN	No value in the audited comparison
Wikidata	κ Boo

Assessment. HIP 69481 is κ1 Boo, currently identified as the B member of 17 Boo in SIMBAD. BSC distinguishes HR 5328 and HR 5329. The Wikidata bare κ Boo is broader and cannot determine which component the HIP marks.

Retention / reporting recommendation. Preserve κ1 for the specific object and κ as a qualified family/system label. Include the component distinction in a report when relevant.

Open question or limit. Do not assume that exponent 1 means the A or brightest component; this case demonstrates otherwise.

Current SIMBAD: * kap01 Boo; ICRS 213.36593380°, 51.78795069°. Kostjuk HD: 124674; HR: 5328.

Evidence: Kostjuk table1 HD 124674; BSC HR 5328–5329; SIMBAD HIP 69481.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

38. HIP 71500 — a Lup

Latin-letter case problem • Report priority: high • Classification confidence: moderate

Audited source	bayer evidence
Kostjuk	a Lup
BSC	No value in the audited comparison
HYG	No value in the audited comparison
WGSN	No value in the audited comparison
Wikidata	A Lup

Assessment. HIP 71500 is Latin a Lup in Kostjuk and SIMBAD, whereas Wikidata has uppercase A Lup. Hoffleit also traces historical reuse of a Lup, so case and identity deserve care. The current HIP identifies a multiple object, HD 128266 / HR 5450. The examined evidence supports lowercase a but does not establish uppercase A as an equivalent historical form.

Retention / reporting recommendation. Preserve the source string A as disputed evidence; keep case-sensitive identifiers. The history of reused Latin letters is reportable separately from this capitalization mismatch.

Open question or limit. Find the upstream source for uppercase A before accepting it as an alias; do not attribute the difference to typography without evidence.

Current SIMBAD: * a Lup; ICRS 219.33398637°, -46.13343919°. Kostjuk HD: 128266; HR: 5450.

Evidence: Hoffleit 1979, *Lupus*, printed p.53; BSC HR 5450; SIMBAD HIP 71500.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hoffleit](#). Full references follow.

39. HIP 71681 — α Cen

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	α Cen

BSC	α 2 Cen
HYG	α 2 Cen
WGSN	α Cen
Wikidata	α 2 Cen

Assessment. HIP 71681 is α Centauri B / HD 128621 / HR 5460, named Toliman. BSC and SIMBAD retain α 2, while Kostjuk and the WGSN Bayer column give bare α . The proper-name target remains B even where the Bayer string is unsuffixed.

Retention / reporting recommendation. Retain α 2 or the explicit B component identity for resolution, and preserve bare α as catalogue wording. A report should distinguish Toliman from the A star.

Open question or limit. Do not interpret WGSN's abbreviated Bayer field as naming the whole system Toliman.

Current SIMBAD: * alf Cen B; ICRS 219.89609629°, -60.83752757°. Kostjuk HD: 128621; HR: 5460. WGSN name: Toliman.

Evidence: Kostjuk table1 HD 128621; BSC HR 5460; WGSN Toliman row; SIMBAD HIP 71681.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

40. HIP 71683 — α Cen

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	α Cen
BSC	α 1 Cen
HYG	α 1 Cen
WGSN	α Cen
Wikidata	α Cen, α 1 Cen

Assessment. HIP 71683 is α Centauri A / HD 128620 / HR 5459, named Rigil Kentaurus. BSC and SIMBAD identify α 1; the bare α strings in other sources reflect broader naming practice. This is a different HIP from Toliman, not a duplicate to be eliminated.

Retention / reporting recommendation. Preserve the A/B distinction and connect the two objects through a system identifier for reports.

Open question or limit. Historical proper-name reuse in BSC is not sufficient to override the modern name-to-component targets.

Current SIMBAD: * alf Cen A; ICRS 219.90205833°, -60.83399269°. Kostjuk HD: 128620; HR: 5459. WGSN name: Rigil Kentaurus.

Evidence: Kostjuk table1 HD 128620; BSC HR 5459; WGSN Rigil Kentaurus row; SIMBAD HIP 71683.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

41. HIP 71762 — π 1 Boo

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	$\pi 1$ Boo
BSC	$\pi 1$ Boo
HYG	$\pi 1$ Boo
WGSN	No value in the audited comparison
Wikidata	π Boo

Assessment. HIP 71762 / HD 129174 is $\pi 1$ Boo, also bare π Boo in SIMBAD. BSC distinguishes HR 5475 from HR 5476 and notes blended light. Kostjuk additionally preserves a competing numbered attribution from one source.

Retention / reporting recommendation. Retain the specific $\pi 1$ designation and document the unsuffixed and competing forms with provenance.

Open question or limit. A coordinate/component mapping is needed before treating the source's $\pi 2$ alternative as a safe resolver alias.

Current SIMBAD: * $\pi 0.1$ Boo; ICRS 220.18149547°, 16.41832666°. Kostjuk HD: 129174; HR: 5475.

Evidence: Kostjuk table1 HD 129174; BSC HR 5475–5476.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

42. HIP 71974 — 4 Lib

Historical shared constellation • Report priority: high • Classification confidence: high

Audited source	flam evidence
Kostjuk	4 Lib
BSC	4 Lib
HYG	4 Lib
WGSN	No value in the audited comparison
Wikidata	4 Lib, 53 Hya

Assessment. 4 Lib and 53 Hya refer to HIP 71974 / HD 129433. The duplicate addendum, Wagman's study and SIMBAD agree on the pairing. The old Hydra number is historically valid evidence even though the modern object is catalogued in Libra.

Retention / reporting recommendation. Retain 53 Hya as a historical alias and describe the duplicate once in an optional report.

Open question or limit. The duplicate's historical cause should not be reduced automatically to a modern boundary reassignment.

Current SIMBAD: * 4 Lib; ICRS 220.80646373°, -24.99775323°. Kostjuk HD: 129433; HR: 5484.

Evidence: Kostjuk addendum HD 129433; Wagman 1987, Hydra, printed p.217; Flamsteed reconstruction.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wagman](#); [flamsteed](#). Full references follow.

43. HIP 72571 — 58 Hya

Audited source	flam evidence
Kostjuk	58 Hya
BSC	58 Hya
HYG	58 Hya
WGSN	No value in the audited comparison
Wikidata	58 Hya, 6 Lib

Assessment. 58 Hya and 6 Lib identify HIP 72571 / HD 130694. The addendum marks the duplicate, and Wagman and SIMBAD corroborate it. This is the reciprocal-looking partner of the 4 Lib / 53 Hya case, but it is a separate star and historical event.

Retention / reporting recommendation. Preserve 6 Lib as a historical alias; a chart with both objects could present a combined discussion of the Libra-Hydra lists.

Open question or limit. Keep the two HIP identities separate even if a report groups their related catalogue history.

Current SIMBAD: * E Hya; ICRS 222.57211788°, -27.96036123°. Kostjuk HD: 130694; HR: 5526. WGSN name: Solitaire.

Evidence: Kostjuk addendum HD 130694; Wagman 1987, Libra, printed p.218.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wagman](#). Full references follow.

44. HIP 74392 — ι Lib

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ι Lib
BSC	ι1 Lib
HYG	ι1 Lib
WGSN	No value in the audited comparison
Wikidata	ι1 Lib

Assessment. 24 Lib / HD 134759 is ι Lib or ι1 Lib in Kostjuk and SIMBAD. BSC records a complex component structure. The bare/suffixed disagreement is real catalogue wording, while the superscript should not be confused with the multiple system's A/P component letters.

Retention / reporting recommendation. Retain both forms and maintain a separate component-scope field. A multiplicity note is possible after modern-source validation.

Open question or limit. BSC's historical orbit and proposed component count are not current astrophysical measurements.

Current SIMBAD: * ιot Lib; ICRS 228.05537606°, -19.79171089°. Kostjuk HD: 134759; HR: 5652.

Evidence: Kostjuk table1 HD 134759; BSC HR 5652.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

45. HIP 75411 — μ 1 Boo

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	μ 1 Boo
BSC	μ 1 Boo
HYG	μ 1 Boo
WGSN	μ Boo
Wikidata	μ Boo

Assessment. Alkalurops at HIP 75411 / HD 137391 is μ 1 Boo; SIMBAD also retains μ Boo. BSC separates HR 5733 from the close double represented by HR 5734. The bare WGSN/Wikidata expression does not imply that every component has the proper name.

Retention / reporting recommendation. Preserve component-qualified μ 1 and the broader alias; explain the visible multiple object in an optional report.

Open question or limit. Use a modern component crosswalk before assigning names or component letters to the secondary pair.

Current SIMBAD: * mu.01 Boo; ICRS 231.12263112°, 37.37715592°. Kostjuk HD: 137391; HR: 5733. WGSN name: Alkalurops.

Evidence: Kostjuk table1 HD 137391; BSC HR 5733–5734; WGSN Alkalurops row.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

46. HIP 76126 — ζ Lib

Superscript convention • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ζ Lib
BSC	ζ 4 Lib
HYG	ζ 4 Lib
WGSN	No value in the audited comparison
Wikidata	ζ 4 Lib

Assessment. 35 Lib / HD 138485 is ζ Lib or ζ 4 Lib. Kostjuk explicitly preserves both, and BSC says the star is usually called simply ζ Lib. SIMBAD retains both identifiers. The expanded superscript distinguishes a historical letter group, not automatically a fourth physical component.

Retention / reporting recommendation. Preserve ζ 4 as a disambiguating form and ζ as a documented common alias.

Open question or limit. Do not infer a four-star bound system from the numeral 4.

Current SIMBAD: * zet04 Lib; ICRS 233.23008856°, -16.85284478°. Kostjuk HD: 138485; HR: 5764.

Evidence: Kostjuk table1 HD 138485; BSC HR 5764 name note.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

47. HIP 76669 — ζ1 CrB / ζ2 CrB

System and component scope • Report priority: high • Classification confidence: moderate

Audited source	bayer evidence
Kostjuk	ζ1 CrB, ζ2 CrB
BSC	ζ1 CrB
HYG	ζ1 CrB
WGSN	No value in the audited comparison
Wikidata	ζ CrB, ζ2 CrB

Assessment. HIP 76669 collects more than one source component row. Kostjuk has ζ1 and ζ2 for the HIP; BSC distinguishes HR 5833 and 5834. Current SIMBAD associates HIP 76669 with 7 CrB A / ζ2 CrB / HD 139892 and also a combined HD identifier. The ledger's ζ1 versus ζ2 comparison is therefore entangled with the cross-index's resolution.

Retention / reporting recommendation. Preserve all evidence with HD, HR and component targets. Report the double without pretending the competing exponents are aliases of one resolved component.

Open question or limit. The HIP-to-component crosswalk needs reconciliation before a canonical suffix or name resolver is approved.

Current SIMBAD: * 7 CrB A; ICRS 234.84447018°, 36.63581106°. Kostjuk HD: 139891, 139892; HR: 5833, 5834.

Evidence: Kostjuk HD 139891–139892 rows and table1 HD 139892; BSC HR 5833–5834; SIMBAD HIP 76669.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

48. HIP 80463 — 24 Her

Historical shared constellation • Report priority: high • Classification confidence: high

Audited source	flam evidence
Kostjuk	24 Her
BSC	24 Her
HYG	24 Her
WGSN	No value in the audited comparison
Wikidata	24 Her, 51 Ser

Assessment. 24 Her / ω Her, Cujam, and 51 Ser identify HIP 80463 / HD 148112. The duplicate addendum, Wagman and SIMBAD agree. The historical Serpens number belongs to the same object rather than a second chart point.

Retention / reporting recommendation. Preserve 51 Ser as a historical alias and use Cujam's identity to anchor any report paragraph.

Open question or limit. A fuller account of the proper name's cultural origin would need its own source review.

Current SIMBAD: * ome Her; ICRS 246.35397781°, 14.03326724°. Kostjuk HD: 148112; HR: 6117. WGSN name: Cujam.

Evidence: Kostjuk addendum HD 148112; BSC HR 6117; Wagman 1987, Serpens, printed p.221.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wagman](#). Full references follow.

49. HIP 81007 — 28 Her

Historical shared constellation • Report priority: high • Classification confidence: high

Audited source	flam evidence
Kostjuk	28 Her
BSC	28 Her
HYG	28 Her
WGSN	No value in the audited comparison
Wikidata	11 Oph, 28 Her

Assessment. 28 Her / Latin n Her and 11 Oph identify HIP 81007 / HD 149121. The addendum explicitly marks the duplicate and SIMBAD retains both numbers. This is historical enumeration across constellation lists, not an erroneous extra HIP.

Retention / reporting recommendation. Preserve 11 Oph as a historical alias. A report can place it alongside the other Hercules–Ophiuchus duplicates.

Open question or limit. The historical duplication is established; authorship and date beyond the numbered-list reconstruction need direct catalogue citation.

Current SIMBAD: * n Her; ICRS 248.14868927°, 5.52121994°. Kostjuk HD: 149121; HR: 6158.

Evidence: Kostjuk addendum HD 149121; Flamsteed reconstruction; SIMBAD HIP 81007.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [flamsteed](#). Full references follow.

50. HIP 81710 — η 1 TrA

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	η 1 TrA
BSC	η 1 TrA
HYG	η 1 TrA
WGSN	No value in the audited comparison
Wikidata	η TrA

Assessment. HIP 81710 / HD 149671 is η 1 TrA in BSC and SIMBAD, but η TrA in the Becvar-linked alternative. Kostjuk explicitly records both. Hoffleit's Triangulum Australe entry likewise treats the exponent as a naming distinction with a faint counterpart.

Retention / reporting recommendation. Preserve the numbered and bare forms without treating the latter as a failed identity.

Open question or limit. A report about the second η object requires its own stable identifier and verification.

Current SIMBAD: * eta01 TrA; ICRS 250.34628230°, -68.29612193°. Kostjuk HD: 149671; HR: 6172.

Evidence: Kostjuk table1 HD 149671; BSC HR 6172; Hoffleit 1979, printed p.64.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hoffleit](#). Full references follow.

51. HIP 82073 — 43 Her

Historical shared constellation • Report priority: high • Classification confidence: high

Audited source	flam evidence
Kostjuk	43 Her
BSC	43 Her
HYG	43 Her
WGSN	No value in the audited comparison
Wikidata	17 Oph, 43 Her

Assessment. 43 Her / Latin i Her and 17 Oph identify HIP 82073 / HD 151217. The duplicate addendum and SIMBAD corroborate the historical pairing. Both numeral designations may be worth preserving even when only one is used for a modern chart label.

Retention / reporting recommendation. Keep 17 Oph as a historical alias and record the common HIP identity for report deduplication.

Open question or limit. Do not conflate Latin i Her with Greek ι Her in a resolver or report.

Current SIMBAD: * i Her; ICRS 251.45788312°, 8.58261352°. Kostjuk HD: 151217; HR: 6228.

Evidence: Kostjuk addendum HD 151217; Flamsteed reconstruction; SIMBAD HIP 82073.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [flamsteed](#). Full references follow.

52. HIP 84345 — α1 Her

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	α1 Her
BSC	α1 Her
HYG	α1 Her
WGSN	α1 Her
Wikidata	α Her

Assessment. Rasalgethi's HIP 84345 is α Her AB in SIMBAD, while HD 156014 / HR 6406 is represented as α1 in the source rows. BSC distinguishes HR 6407, and Kostjuk preserves bare and numbered alternatives. This is another system/component mismatch, not evidence for a second constellation assignment.

Retention / reporting recommendation. Retain the precise component label and the broad system alias separately. A report may discuss the multiple star after current astrophysical verification.

Open question or limit. The proper name's target should follow the explicit WGSN row rather than the unsuffixed string alone.

Current SIMBAD: * alf Her; ICRS 258.66190909°, 14.39034061°. Kostjuk HD: 156014; HR: 6406. WGSN name: Rasalgethi.

Evidence: Kostjuk table1 HD 156014; BSC HR 6406–6407; WGSN Rasalgethi row.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

53. HIP 86614 — ψ Dra

System and component scope • Report priority: high • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ψ Dra
BSC	ψ 1 Dra
HYG	ψ 1 Dra
WGSN	ψ 1 Dra
Wikidata	No value in the audited comparison

Assessment. HIP 86614 is ψ 1 Dra A / HD 162003 / HR 6636, named Dziban in WGSN. Bare ψ is a documented catalogue convention. The missing Wikidata Bayer value is a coverage gap, not an argument against the designation. BSC records a competing historical numbering scheme involving the nearby B component and a separate ψ 2 star.

Retention / reporting recommendation. Preserve ψ 1 Dra A and the bare catalogue alias. Report the naming ambiguity jointly with HIP 86620, without assigning Dziban to both.

Open question or limit. The local double's A/B labels must be kept distinct from the wider ψ 1/ ψ 2 numbering.

Current SIMBAD: * psi01 Dra A; ICRS 265.48359321°, 72.14886878°. Kostjuk HD: 162003; HR: 6636. WGSN name: Dziban.

Evidence: Kostjuk table1 HD 162003; BSC HR 6636–6637; WGSN Dziban row; SIMBAD HIP 86614.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

54. HIP 86620 — ψ Dra

System and component scope • Report priority: high • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ψ Dra
BSC	ψ 1 Dra
HYG	ψ 1 Dra
WGSN	No value in the audited comparison
Wikidata	No value in the audited comparison

Assessment. HIP 86620 is ψ 1 Dra B / HD 162004 / HR 6637 in SIMBAD. BSC explicitly says some catalogues number HR 6636/6637 as ψ 1/ ψ 2, whereas most treat both as ψ 1 and reserve ψ 2 for HR

6725. Kostjuk preserves both traditions for this HD. The missing Wikidata value is secondary to that genuine historical ambiguity.

Retention / reporting recommendation. Retain the explicit B component identity and both historical naming schemes with provenance. Do not rename this HIP ψ 2 merely because it is the second nearby HIP.

Open question or limit. Check the separate HR 6725 object whenever resolving an unqualified ψ 2 Dra query.

Current SIMBAD: * psi01 Dra B; ICRS 265.49209682°, 72.15690156°. Kostjuk HD: 162004; HR: 6637.

Evidence: BSC HR 6637 name note; Kostjuk table1 HD 162004; SIMBAD HIP 86620.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

55. HIP 86974 — μ Her

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	μ Her
BSC	μ Her
HYG	μ Her
WGSN	No value in the audited comparison
Wikidata	μ 1 Her

Assessment. HIP 86974 / HD 161797 is μ Her, also μ 1 Her and μ Her A in SIMBAD. BSC records a multiple system. Wikidata's extra 1 is compatible with a component-qualified tradition rather than necessarily a fabricated designation, but it is not documented in Kostjuk's alternative table.

Retention / reporting recommendation. Preserve the system and primary aliases with scope; distinguish exponent numbering from A/B/C component letters.

Open question or limit. The specific historical source for Wikidata's μ 1 statement should be attached before a historical chronology is written.

Current SIMBAD: * mu.01 Her; ICRS 266.61478040°, 27.72071138°. Kostjuk HD: 161797; HR: 6623.

Evidence: BSC HR 6623; SIMBAD HIP 86974.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

56. HIP 87314 — ν 1 Ara

Competing letter traditions • Report priority: high • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ν 1 Ara
BSC	No value in the audited comparison
HYG	No value in the audited comparison
WGSN	No value in the audited comparison
Wikidata	ν 1 Ara

Assessment. HIP 87314 / HD 161783 / HR 6622 has υ1 Ara and v1 Ara documented in the same source evidence. BSC explicitly notes the competing upsilon/nu usages; Hoffleit traces older naming practice. Current SIMBAD's main object is V539 Ara and the returned identifier set lacks either Bayer form. This is a historical letter dispute, not proof that Wikidata confused two Unicode characters.

Retention / reporting recommendation. Preserve both Greek-letter forms with provenance and use the stable HIP/HD identity. A report can explain the competing nomenclature.

Open question or limit. Determine the preferred current label separately; SIMBAD's omission is not a ruling that neither historical form existed.

Current SIMBAD: V* V539 Ara; ICRS 267.61830417°, -53.61240688°. Kostjuk HD: 161783; HR: 6622.

Evidence: BSC HR 6622 name note; Kostjuk table1 HD 161783; Hoffleit 1979, Ara, printed p.44.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hoffleit](#). Full references follow.

57. HIP 87379 — υ2 Ara

Competing letter traditions • Report priority: high • Classification confidence: high

Audited source	bayer evidence
Kostjuk	υ2 Ara
BSC	No value in the audited comparison
HYG	No value in the audited comparison
WGSN	No value in the audited comparison
Wikidata	υ2 Ara

Assessment. HIP 87379 / HD 161917 / HR 6632 has the parallel υ2 Ara / v2 Ara history. BSC explicitly records both, and SIMBAD currently retains both identifiers. It is a useful paired example with HIP 87314, showing that the discrepancy follows a naming tradition rather than a lone accidental character change.

Retention / reporting recommendation. Preserve v2 and υ2 as qualified aliases and group the two Ara cases in a report.

Open question or limit. Do not infer the original reason for nu replacing upsilon without an atlas or catalogue-specific account.

Current SIMBAD: * nu.02 Ara; ICRS 267.79602211°, -53.13026904°. Kostjuk HD: 161917; HR: 6632.

Evidence: BSC HR 6632 name note; Kostjuk table1 HD 161917; Hoffleit 1979, Ara, printed p.44.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hoffleit](#). Full references follow.

58. HIP 88635 — γ Sgr

Superscript convention • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	γ Sgr
BSC	γ2 Sgr

HYG	γ 2 Sgr
WGSN	γ 2 Sgr
Wikidata	γ 2 Sgr

Assessment. Alnasl / 10 Sgr / HD 165135 is γ Sgr or γ 2 Sgr. Kostjuk and SIMBAD preserve both; BSC additionally mentions γ 1 or γ 2 usage and points to HR 6742. The bare source form is therefore compatible with documented numbering variation.

Retention / reporting recommendation. Preserve bare γ and γ 2, with the WGSN name tied to the identified HIP. A report may explain the historical group if relevant.

Open question or limit. Do not turn γ 1 into a resolver alias of this HIP without checking the distinct HR 6742 star and the source's meaning.

Current SIMBAD: * gam02 Sgr; ICRS 271.45203374°, -30.42408985°. Kostjuk HD: 165135; HR: 6746. WGSN name: Alnasl.

Evidence: Kostjuk table1 HD 165135; BSC HR 6746 name note; WGSN Alnasl row.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

59. HIP 89153 — 11 Sgr

Identity or cross-index problem • Report priority: high • Classification confidence: high

Audited source	flam evidence
Kostjuk	11 Sgr
BSC	11 Sgr
HYG	1 Sgr
WGSN	No value in the audited comparison
Wikidata	1 Sgr

Assessment. HIP 89153 / HR 6801 is 1 Sgr in current SIMBAD and the Flamsteed reconstruction. The 11 Sgr used by Kostjuk and BSC has a documented history: Hoffleit advocated it in 1979, but Wagman explicitly corrected that interpretation in 1987. The actual 11 Sgr is HD 165921, a separate faint star. This is both a likely misidentification and a valuable history of correction.

Retention / reporting recommendation. Prefer preserving 11 Sgr as a disputed historical attribution, not an ordinary interchangeable alias. Report the catalogue correction story with attribution.

Open question or limit. Fernando's label decision remains pending; an original Baily/Flamsteed transcription would further strengthen the historical account.

Current SIMBAD: * 1 Sgr; ICRS 272.93055316°, -23.70123160°. Kostjuk HD: 166464; HR: 6801.

Evidence: Hoffleit 1979, printed p.62; Wagman 1987, Sagittarius, printed p.221; Flamsteed reconstruction.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hoffleit](#); [wagman](#); [flamsteed](#). Full references follow.

60. HIP 90968 — κ 2 CrA

Identity or cross-index problem • Report priority: high • Classification confidence: moderate

Audited source	bayer evidence
Kostjuk	$\kappa 2$ CrA
BSC	$\kappa 1$ CrA
HYG	$\kappa 1$ CrA
WGSN	No value in the audited comparison
Wikidata	$\kappa 2$ CrA

Assessment. HIP 90968 has $\kappa 2$ CrA in Kostjuk and SIMBAD. The apparent BSC $\kappa 1$ disagreement results from the HD crosswalk: Kostjuk and the original Hipparcos table assign HD 170868, but current SIMBAD assigns HD 170867 / HR 6953. BSC's HR 6953 position matches this southern, brighter star and calls it $\kappa 2$. This is a deeper identity-association conflict, not simply a superscript disagreement at fixed coordinates.

Retention / reporting recommendation. Quarantine automatic preference selection for this pair and compare coordinates, brightness, HD and HR together. Preserve all source assertions with their exact targets.

Open question or limit. Resolve whether the original HIP HD field or another cross-index should be corrected; no catalogue repair is applied in this review.

Current SIMBAD: * kap02 CrA; ICRS 278.34638041°, -38.72598312°. Kostjuk HD: 170868; HR: 6952.

Evidence: Kostjuk rows HD 170867–170868; BSC HR 6952–6953; Hipparcos I/239; SIMBAD HIP 90968–90969.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hipparcos](#). Full references follow.

61. HIP 90969 — $\kappa 1$ CrA

Identity or cross-index problem • Report priority: high • Classification confidence: moderate

Audited source	bayer evidence
Kostjuk	$\kappa 1$ CrA
BSC	$\kappa 2$ CrA
HYG	$\kappa 2$ CrA
WGSN	No value in the audited comparison
Wikidata	$\kappa 1$ CrA

Assessment. HIP 90969 is the northern, fainter $\kappa 1$ CrA star in Kostjuk and SIMBAD. Kostjuk and the original Hipparcos table associate it with HD 170867, whereas SIMBAD associates HD 170868 / HR 6952. BSC's $\kappa 1$ label at HR 6952 agrees spatially with this HIP. The ledger's $\kappa 2$ value came through the contested HD association.

Retention / reporting recommendation. Review together with HIP 90968; retain stable positions and explicit source-specific cross-identifications. Do not swap superscripts in isolation.

Open question or limit. The paired HD/HR inconsistency requires a provenance-aware join policy and an explicit curation decision.

Current SIMBAD: * kap01 CrA; ICRS 278.34617048°, -38.72004622°. Kostjuk HD: 170867; HR: 6953.

Evidence: Kostjuk rows HD 170867–170868; BSC HR 6952–6953; Hipparcos I/239; SIMBAD HIP 90968–90969.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hipparcos](#). Full references follow.

62. HIP 91926 — ε2 Lyr

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ε2 Lyr
BSC	ε2 Lyr
HYG	ε2 Lyr
WGSN	No value in the audited comparison
Wikidata	ε Lyr

Assessment. HIP 91926 is ε2 Lyr, the CD pair within the wider Double Double. SIMBAD identifies a CD system and preserves ε2; Wikidata's bare ε Lyr is broader. BSC's HR 7053 is a component row rather than the entire four-star complex.

Retention / reporting recommendation. Preserve ε2 for this HIP and ε Lyr as a broader family/system designation. A report can explain the hierarchy without merging the ε1 HIP.

Open question or limit. Validate modern subcomponent details before quoting separations, periods or component counts from old notes.

Current SIMBAD: * eps02 Lyr; ICRS 281.09491902°, 39.61271808°. Kostjuk HD: 173607; HR: 7053.

Evidence: BSC HR 7051–7054; SIMBAD HIP 91926.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

63. HIP 92405 — υ Lyr

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	υ Lyr
BSC	υ2 Lyr
HYG	υ2 Lyr
WGSN	No value in the audited comparison
Wikidata	υ2 Lyr

Assessment. 9 Lyr / HD 174602 is υ Lyr or υ2 Lyr. BSC explicitly says the star is usually called υ and identifies υ1 as HR 7100. Kostjuk and SIMBAD retain both forms on the same object.

Retention / reporting recommendation. Preserve υ2 as the disambiguating form and υ as a documented usual alias.

Open question or limit. The numeral is not evidence that υ1 and υ2 form a physical binary.

Current SIMBAD: * nu. Lyr; ICRS 282.47048883°, 32.55106029°. Kostjuk HD: 174602; HR: 7102.

Evidence: BSC HR 7102 name note; Kostjuk table1 HD 174602.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

64. HIP 95853 — ι Cyg

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ι Cyg
BSC	ι2 Cyg
HYG	ι2 Cyg
WGSN	No value in the audited comparison
Wikidata	ι2 Cyg

Assessment. 10 Cyg / HD 184006 has ι Cyg and ι2 Cyg documented by Kostjuk and SIMBAD. The compared sources agree on identity and differ in whether they retain the letter-group exponent.

Retention / reporting recommendation. Preserve both aliases; no prominent error or historical-warning report is necessary by default.

Open question or limit. A more detailed historical origin of the exponent has not been reconstructed.

Current SIMBAD: * iot Cyg; ICRS 292.42649261°, 51.72977792°. Kostjuk HD: 184006; HR: 7420.

Evidence: Kostjuk table1 HD 184006; BSC HR 7420; Hoffleit 1979, printed p.49.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hoffleit](#). Full references follow.

65. HIP 95947 — β1 Cyg

System and component scope • Report priority: high • Classification confidence: high

Audited source	bayer evidence
Kostjuk	β1 Cyg
BSC	β1 Cyg
HYG	β1 Cyg
WGSN	β Cyg
Wikidata	No value in the audited comparison

Assessment. HIP 95947 is Albireo's β1 Cyg / A object in SIMBAD, while WGSN uses β Cyg in its name row. The missing Wikidata Bayer field is a coverage issue. The proper name itself has a documented textual-transmission history in the WGSN origin entry, making this object interesting independently of its audit discrepancy.

Retention / reporting recommendation. Keep β1/A distinct from the other visual star and retain Albireo's historical name story as a separately sourced report topic.

Open question or limit. Do not repeat BSC's old claim or orbit for the wide pair as current fact; any physical-association statement requires modern research.

Current SIMBAD: * bet01 Cyg; ICRS 292.68031501°, 27.95967363°. Kostjuk HD: 183912; HR: 7417. WGSN name: Albireo.

Evidence: Kostjuk table1 HD 183912; BSC HR 7417–7418; WGSN Albireo origin and target row.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

66. HIP 98162 — b1 Sgr

Superscript convention • Report priority: high • Classification confidence: moderate

Audited source	bayer evidence
Kostjuk	b1 Sgr
BSC	No value in the audited comparison
HYG	No value in the audited comparison
WGSN	No value in the audited comparison
Wikidata	b Sgr

Assessment. 59 Sgr / HD 188603 is Latin b Sgr or b1 Sgr in Kostjuk's alternatives; SIMBAD uses b. BSC also records Terebellum as a name shared with three other HR stars. This makes group-name scope valuable for reporting, even though the letter discrepancy itself is ordinary catalogue convention.

Retention / reporting recommendation. Retain b/b1 with provenance. Preserve Terebellum as an asterism or historical group name rather than assigning it exclusively to this HIP.

Open question or limit. Hoffleit did not establish the source of the exponent; verify a modern group-membership crosswalk before reporting all four stars.

Current SIMBAD: * b Sgr; ICRS 299.23680287°, -27.16989701°. Kostjuk HD: 188603; HR: 7604.

Evidence: Kostjuk table1 HD 188603; BSC HR 7604 name note; Hoffleit 1979, printed p.62.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hoffleit](#). Full references follow.

67. HIP 99572 — ξ Cap

Superscript convention • Report priority: low • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ξ Cap
BSC	ξ2 Cap
HYG	ξ2 Cap
WGSN	No value in the audited comparison
Wikidata	ξ Cap, ξ2 Cap

Assessment. 2 Cap / HD 191862 is ξ Cap or ξ2 Cap. Kostjuk explicitly documents both and SIMBAD retains both; Wikidata's two values consequently preserve a real catalogue convention rather than two conflicting stellar identities.

Retention / reporting recommendation. Preserve both forms and distinguish display abbreviation from object resolution.

Open question or limit. A physical relationship with another ξ star is not implied by the exponent.

Current SIMBAD: * ksi Cap; ICRS 303.10779404°, -12.61749753°. Kostjuk HD: 191862; HR: 7715.

Evidence: Kostjuk table1 HD 191862; BSC HR 7715.
Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

68. HIP 100345 — β Cap

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	β Cap
BSC	β Cap
HYG	β Cap
WGSN	$\beta 1$ Cap
Wikidata	No value in the audited comparison

Assessment. HIP 100345 is $\beta 1$ Cap / Dabih Major in SIMBAD. BSC's main label is bare β , but its note explicitly distinguishes $\beta 1$ Major and $\beta 2$ Minor. WGSN's $\beta 1$ therefore adds legitimate resolution, and the missing Wikidata value is a coverage gap.

Retention / reporting recommendation. Preserve the primary/subsystem identity and the bare historical label; connect $\beta 2$ through a system relation for a report.

Open question or limit. The complex hierarchy and current physical measurements need modern sources before an observing narrative is published.

Current SIMBAD: * bet01 Cap; ICRS 305.25277749°, -14.78140760°. Kostjuk HD: 193495; HR: 7776. WGSN name: Dabih.

Evidence: BSC HR 7776 name and double notes; Kostjuk table1 HD 193495; WGSN Dabih row.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

69. HIP 101923 — τ Cap

Superscript convention • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	τ Cap
BSC	τ Cap
HYG	τ Cap
WGSN	No value in the audited comparison
Wikidata	$\tau 2$ Cap

Assessment. 14 Cap / HD 196662 is usually τ Cap, also $\tau 2$ Cap. BSC explicitly identifies $\tau 1$ as 13 Cap / HD 196348, a fainter star in the supplement. Kostjuk and SIMBAD preserve the two forms for HIP 101923. This demonstrates that a missing faint counterpart in BSC's main table can hide a valid numbering distinction.

Retention / reporting recommendation. Retain $\tau 2$ and the usual bare form. A short report can explain the fainter numbered counterpart when relevant.

Open question or limit. Do not infer a physical relation from the τ group, despite multiplicity within this HIP's own object.

Current SIMBAD: * tau Cap; ICRS 309.81799079°, -14.95475977°. Kostjuk HD: 196662; HR: 7889.

Evidence: BSC HR 7889 name note; Kostjuk table1 HD 196662.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

70. HIP 107310 — μ 1 Cyg

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	μ 1 Cyg
BSC	μ 1 Cyg
HYG	μ 1 Cyg
WGSN	No value in the audited comparison
Wikidata	μ Cyg

Assessment. HIP 107310 is μ Cyg AB in SIMBAD; the compared HD 206826 / HR 8309 component row uses μ 1. BSC distinguishes HR 8310, while Kostjuk records both exponents in different source traditions. The ledger's bare μ form is therefore at a broader scope.

Retention / reporting recommendation. Preserve μ at system scope and numbered labels with component/source targets. A report can describe the visual double after current verification.

Open question or limit. Resolve competing μ 1/ μ 2 assignments by coordinates and component IDs before using them as unique aliases.

Current SIMBAD: * mu. Cyg; ICRS 326.03574030°, 28.74262687°. Kostjuk HD: 206826; HR: 8309.

Evidence: Kostjuk table1 HD 206826; BSC HR 8309–8310.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

71. HIP 107382 — c Cap

Superscript convention • Report priority: low • Classification confidence: moderate

Audited source	bayer evidence
Kostjuk	c Cap
BSC	No value in the audited comparison
HYG	No value in the audited comparison
WGSN	No value in the audited comparison
Wikidata	c1 Cap

Assessment. 46 Cap / HD 206834 is Latin c Cap in Kostjuk and SIMBAD, with c1 in a Becvar-linked alternative. BSC gives the Flamsteed number but not the Latin letter in its main name field. The evidence supports a source-specific exponent rather than a separate HIP identity.

Retention / reporting recommendation. Preserve Latin case and both source strings; this is a specialist nomenclature note.

Open question or limit. The first introduction and intended scope of c1 remain untraced.

Current SIMBAD: * c Cap; ICRS 326.25106114°, -9.08242767°. Kostjuk HD: 206834; HR: 8311.

Evidence: Kostjuk table1 HD 206834; BSC HR 8311.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

72. HIP 108478 — κ Ind

Superscript convention • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	κ Ind
BSC	κ1 Ind
HYG	κ1 Ind
WGSN	No value in the audited comparison
Wikidata	κ1 Ind

Assessment. HIP 108478 / HD 208496 is κ1 Ind in BSC and SIMBAD; Kostjuk's bare κ follows the Becvar-linked convention. It must remain distinct from HIP 109081, which is κ2 Ind. The shared bare source label is therefore less specific than the two HIP identities.

Retention / reporting recommendation. Keep κ1 attached to this HIP and retain κ Ind as a coarse catalogue alias. A report may describe the naming pair if both appear.

Open question or limit. No physical association of the two κ stars is asserted by this review.

Current SIMBAD: * kap01 Ind; ICRS 329.62533014°, -59.01214541°. Kostjuk HD: 208496; HR: 8369.

Evidence: Kostjuk table1 HD 208496; BSC HR 8369; SIMBAD HIP 108478.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

73. HIP 109081 — κ Ind

Superscript convention • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	κ Ind
BSC	κ2 Ind
HYG	κ2 Ind
WGSN	No value in the audited comparison
Wikidata	κ2 Ind

Assessment. HIP 109081 / HD 209529 is κ2 Ind in BSC and SIMBAD, also bare κ in the Becvar-linked Kostjuk row. Together with HIP 108478 this shows why a bare designation may intentionally cover a group of distinct stars.

Retention / reporting recommendation. Keep $\kappa 2$ specific and preserve bare κ with a non-unique scope marker.

Open question or limit. An unsuffixed κ Ind query should not silently choose one of the two HIPs.

Current SIMBAD: * kap02 Ind; ICRS 331.46256671°, -59.63607214°. Kostjuk HD: 209529; HR: 8409.

Evidence: Kostjuk table1 HD 209529; BSC HR 8409; SIMBAD HIP 109081.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

74. HIP 109410 — π Peg

Superscript convention • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	π Peg
BSC	$\pi 2$ Peg
HYG	$\pi 2$ Peg
WGSN	$\pi 2$ Peg
Wikidata	$\pi 2$ Peg

Assessment. 29 Peg / HD 210459 is π Peg or $\pi 2$ Peg in Kostjuk and SIMBAD; WGSN retains $\pi 2$ and names this object Neichu in the audited snapshot. The exponent disagreement is routine, while the WGSN cultural naming context offers a separate possible report topic.

Retention / reporting recommendation. Preserve $\pi 2$ and the bare alias. Store Neichu's source and adoption metadata separately from the Bayer preference.

Open question or limit. A cultural-history paragraph needs its underlying asterism sources; the catalogue's brief origin note alone should not become an expansive account.

Current SIMBAD: * pi. Peg; ICRS 332.49684591°, 33.17822233°. Kostjuk HD: 210459; HR: 8454. WGSN name: Neichu.

Evidence: Kostjuk table1 HD 210459; BSC HR 8454; WGSN Neichu row.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [wgsn](#). Full references follow.

75. HIP 110960 — $\zeta 1$ Aqr

System and component scope • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	$\zeta 1$ Aqr
BSC	$\zeta 1$ Aqr
HYG	$\zeta 1$ Aqr
WGSN	No value in the audited comparison
Wikidata	ζ Aqr

Assessment. HIP 110960 is ζ Aqr AB in SIMBAD. The compared component HD 213051 / HR 8558 has $\zeta 1$ in BSC, and Kostjuk records a competing $\zeta 2$ source attribution. BSC distinguishes HR 8559. Bare ζ therefore labels a broader object and cannot adjudicate the numbered component

disagreement.

Retention / reporting recommendation. Preserve the system alias and the competing component forms with provenance. Require a coordinate/component crosswalk for resolution.

Open question or limit. Do not equate ζ1 with A across every historical source without checking the convention.

Current SIMBAD: * zet Aqr; ICRS 337.20794520°, -0.01994285°. Kostjuk HD: 213051; HR: 8558.

Evidence: Kostjuk table1 HD 213051; BSC HR 8558–8559; SIMBAD HIP 110960.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

76. HIP 111056 — ρ Cep

Superscript convention • Report priority: medium • Classification confidence: high

Audited source	bayer evidence
Kostjuk	ρ Cep
BSC	ρ2 Cep
HYG	ρ2 Cep
WGSN	No value in the audited comparison
Wikidata	ρ2 Cep

Assessment. 29 Cep / HD 213798 is ρ Cep or ρ2 Cep in Kostjuk and SIMBAD. BSC retains a historical traditional-name form, Al Kalb al Rai, in its notes. The Bayer disagreement is ordinary numbering convention; the traditional name is separate historical evidence, not necessarily a modern standardized proper name.

Retention / reporting recommendation. Preserve both Bayer forms; retain the BSC name with historical status and its source.

Open question or limit. Verify cultural origin and modern name status before promoting the old BSC spelling to a report headline.

Current SIMBAD: * rho Cep; ICRS 337.47074152°, 78.82428668°. Kostjuk HD: 213798; HR: 8591.

Evidence: Kostjuk table1 HD 213798; BSC HR 8591 name note; Hoffleit 1979, printed p.48.

Sources: [kostjuk](#); [bsc](#); [simbad](#); [hoffleit](#). Full references follow.

77. HIP 112211 — g Aqr

Superscript convention • Report priority: low • Classification confidence: moderate

Audited source	bayer evidence
Kostjuk	g Aqr
BSC	No value in the audited comparison
HYG	No value in the audited comparison
WGSN	No value in the audited comparison
Wikidata	g1 Aqr

Assessment. 66 Aqr / HD 215167 is Latin g Aqr in Kostjuk and SIMBAD, with g1 in the Becvar-linked alternative row. BSC's main name field does not include the Latin label, so its blank extracted Bayer value is not contradictory evidence.

Retention / reporting recommendation. Preserve case and the source-specific exponent. Use the Flamsteed or stable HIP identity to disambiguate retrieval.

Open question or limit. The exponent's original introduction has not been established; this is not yet a distinctive report story.

Current SIMBAD: * g Aqr; ICRS 340.89680519°, -18.83037508°. Kostjuk HD: 215167; HR: 8649.

Evidence: Kostjuk table1 HD 215167; BSC HR 8649.

Sources: [kostjuk](#); [bsc](#); [simbad](#). Full references follow.

Data contract for an optional chart report

1. Keep preferred display labels separate from historical aliases, disputed attributions and report topics. A valid historical alias can remain reportable even when never selected for a chart label.
2. Store each assertion with source, source row or page, epoch/edition when known, scope (star/component/subsystem/system/asterism), validity status, confidence and open questions.
3. Use HIP for the plotted object, plus HD/HR/WDS/component relations where HIP represents a blend. A report topic may target multiple HIP objects or a wider system.
4. Generate a requested report from the objects actually rendered after field/horizon/magnitude selection. Label visibility should be a separate eligibility option; an unlabelled plotted object can still be interesting.
5. Deduplicate report topics by event/system identifier, preserve links to each relevant plotted HIP, and explain when additional components are outside the chart.
6. Offer concise historical notes by default within the optional report, with observing/astrophysical detail and catalogue disputes as separately selectable categories.
7. Resolve unstable cross-identifications before they enter the automatic name resolver. Unsupported or erroneous aliases can be retained as evidence while remaining ineligible for ordinary query resolution.
8. Retain citations and editorial/research status in the topic data. Chart-generation time should use reviewed local material, without turning fresh unsupervised database assertions into narrative facts.
9. A later implementation should first add a curated topic dataset and validate HIP/component targets, then add optional report output and CLI/TOML parity. This dossier proposes the data contract only.

Bibliography and source distinctions

kostjuk: Kostjuk (2004), IV/27A cross-index

<https://cdsarc.cds.unistra.fr/ftp/IV/27A/ReadMe>

bsc: Hoffleit & Warren, Bright Star Catalogue, 5th revised edition

<https://cdsarc.cds.unistra.fr/ftp/V/50/ReadMe>

simbad: SIMBAD, CDS; live TAP query for all 77 HIP identifiers

<https://simbad.cds.unistra.fr/simbad/sim-tap>

bayer: Johann Bayer, Uranometria, catalogued 1603 copy; ETH-Bibliothek Zürich, Rar 8931

<https://doi.org/10.3931/e-rara-309>

Original tables inspected visually: Auriga XII, leaf 46; Pegasus XIX, leaf 67; Andromeda XX, leaf 70. Leaf numbers are digital PDF positions, not original pagination.

hoffleit: D. Hoffleit (1979), Discordances in Star Designations, Bulletin CDS 17, pp.38-65

<https://articles.adsabs.harvard.edu/pdf/1979BICDS..17...38H>

Downloaded scan read using OCR; targeted historical readings checked against catalogue notes. OCR Greek letters and table alignments must not be treated as an authoritative transcription.

wagman: M. Wagman (1987), Flamsteed's Missing Stars, Journal for the History of Astronomy 18, pp.209-223

<https://doi.org/10.1177/002182868701800305>

flamsteed: Pierre Barbier, reconstructed Flamsteed identification table; independent reconstruction informed by historical catalogues and Wagman

<https://pbarbier.com/flamsteed/flamsteed-ident.html>

A modern reconstruction, not an original Flamsteed manuscript. Used as an independent identity check, particularly 8/9 Sex and 1/11 Sgr.

wgsn: IAU WGSN Catalog of Star Names; audited snapshot

<https://exopla.net/star-names/modern-iau-star-names>

Proper-name targets distinguished from the preliminary Naked Eye Catalogue. NEC is explicitly preliminary and not used as an identity arbiter.

hipparcos: ESA (1997), Hipparcos Main Catalogue, I/239; κ CrA rows

https://vizier.cds.unistra.fr/viz-bin/asu-tsv?-source=I/239/hip_main&HIP=90968,90969&-out=HIP,HD,RAICRS,DEICRS&-out.max=10

Source hashes and exact audited row/statement identifiers are retained in the accompanying JSON. Research recommendations do not populate Fernando's decision or Selman fields.